

PROJECT MODULE: TASK 1 CONFIGURATION REPORT

Target Infrastructure: Ubuntu 24.04 LTS Instance on Microsoft Azure

1. Scope & System Objective

The objective of this module is to provision a cloud enterprise virtual machine instance and transition its core authentication mechanics from standard password interaction patterns to restricted asymmetric cryptographic key-pair validations. This configuration strictly eliminates brute-force vulnerability points across public networks.

2. Implementation Execution Sequence

1. **Cloud VM Provisioning** : Initialized a clean Ubuntu Server instance node leveraging the Microsoft Azure compute engine platform.
2. **Key-Pair Injection** : Attached localized asymmetric identity public keys directly during system instantiation parameters.
3. **Hardening OpenSSH Directives** : Before modifying core configurations, an isolated static backup copy of the master daemon rules configuration was created:

```
sudo cp /etc/ssh/sshd_config /etc/ssh/sshd_config.bak
```

Accessed the configuration file via superuser privilege boundaries:

```
sudo nano /etc/ssh/sshd_config
```

Modified targeted parameters inside the configuration data blocks to explicitly reject basic credentials requests:

Plaintext

```
PubkeyAuthentication yes
```

```
PasswordAuthentication no
```

4. **Daemon Service Reload** : Refreshed live running platform systems to activate security transformations immediately:

Bash

```
sudo systemctl restart ssh
```

3. Verification & Validation Metrics

- **Authorized Credentials Path** : Logging in with the matching local private token successfully bypasses traditional password prompts.
- **Enforced Isolation Protection** : Remote queries initialized without validation certificates result in direct failure barriers:

Plaintext

```
$ ssh azureuser@<Azure-VM-Public-IP>
```

azureuser@<Azure-VM-Public-IP>: Permission denied (publickey).

```
azureuser@prasad-vm:~$ ssh azureuser@4.205.72.196
The authenticity of host '4.205.72.196 (4.205.72.196)' can't be established.
ED25519 key fingerprint is SHA256:cUyvsN9t9G909B3hvpI6990e30EaEIYsbmQHtf+80HQ.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '4.205.72.196' (ED25519) to the list of known hosts.
azureuser@4.205.72.196: Permission denied (publickey).
azureuser@prasad-vm:~$
```

